

Smart IoT-Based Fault Detection in Underground Cables for Enhanced Infrastructure Reliability

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Abstract—Since the electricity system grid has developed; underground wires have been employed extensively. Because of the subsurface environment, deterioration, and rodent activity, numerous problems might arise with underground lines. Excavating the entire line is necessary to check for cable flaws, locating the source of the fault is challenging. Only that region needs to be excavated in order to find the cause of the fault, as the person who are going to repair are aware precisely which section is defective. As a result, it enables quicker maintenance of subterranean cable systems and saves a significant amount of money. Finding the underground cable faults' distance in kilometers from the base station is the goal of this study.

Index Terms—underground, Fault, Detect, Money, maintenance.

I. INTRODUCTION

In this study, proposed an Arduino-based model for identifying subterranean cable line problems. This research aims to calculate the distance in kilometers from the base station to the underground cable fault. Ohm's law is a basic notion that we employed in this study. The liquid crystal display (LCD) distance indicates when there is a fault in the underground cable. There is now no underground cable that is higher than the previous technique; cables were intended to be positioned above the head until the last ten years [2]. Underground lines are not impacted by inclement weather, including storms, snow, heavy rain, or pollutants. Locating an underground cable fault can be challenging when it arises in underground lines. Intend to precisely determine the fault's location. Since the world has gone digital, the research goal is to determine precisely where it is of the defect in digital form. In many cities, the use of underground cabling systems is becoming more and more common [3]. The precise position of the cable failure makes the repair process more difficult, even when the cause of the malfunction is unclear. Two different kinds of cable faults can be distinguished: Open circuit fault: An open circuit fault is characterized by the absence of current due to the absence of a conducting full loop, or $I=0$. The output and supply voltages are equal in this fault. A short circuit fault is preferable than an exposed track culpability. Tiny track burden: this issue results in zero output voltage but constant current. Two forms of further short circuit faults can be distinguished: Equivalent phase shift and lead current characterize a symmetrical fault [7]. Asymmetrical fault: -

This Fault has phase shifting and current magnitude are not identical by 120 degrees. Terminal method: this approach uses minimal effort to locate faults in underground cables. This technique, which uses the voltage drop as a function of the current variation in the cable's default length, is used to identify the type of circuit. The cable can be represented by a number of resistors, and to speed up the tracking of the buried cable, a DC voltage is applied at one end [1]. The shortcoming is identified by perceiving the voltage dissimilarity in the fault area.

II. LITERATURE SURVEY

The goal of this study is to compute the distance, in kilometers, between an underground cable fault and a base station, and to display the data on an electronic device. Underground cable systems are widely used in metropolitan areas. When a cable malfunctions for any reason, pinpointing the exact location of the issue within the cable makes the repair process for that particular cable more challenging. To overcome this problem, we have proposed a method in which the distance of the fault from the base station in km. The study makes use of the well-known principle of Ohm's law, which states that because resistance is proportionate to distance, current will vary depending on where a cable fault is located when a low DC voltage is applied at the feeder end through a series resistor (cable lines). The voltage between the series resistors varies in response to the resistance, which varies with distance, in the event of a short circuit (line to ground). After that, an ADC receives this voltage and produces accurate digital data. This data is shown in kilometers on an 8051 series microcontroller that has been programmed [4].

Because of the subsurface conditions, deterioration, rodents, etc., numerous problems might arise with underground lines. Additionally, it can be challenging to identify the source of a defect; instead, the complete line must be dug in order to check and correct flaws. Thus, we now suggest cable fault detection over IOT, which locates the precise fault spot over IOT and simplifies the repair process. The repairmen know precisely which section is broken, so they just need to excavate that area to find the root of the problem. This enables quicker maintenance of underground cables while also saving a great deal of time, money, and effort. We make advantage of

Internet of Things technologies to update the monitored fault information online. With the use of a potential divider network placed across the cable, the system is able to identify faults. Every time a fault arises at a location where two lines are shorted together, the resistor network combination produces a particular voltage [12]. The microcontroller intellects this power and informs the user. The information about defect detection that is communicated to the user.

It takes a lot of manpower and resources to locate transmission line faults. This is usually a labor-intensive procedure, and there is a chance that the insulation will be harmed during the digging of the cable. This paper automates the defect location and detection procedure, offering a straightforward and secure substitute. This study uses a series resistor to apply a low DC voltage at the feeder end while utilizing the straightforward idea of Ohm's law [17].

The current in the case of a short circuit, such as LL, 3L, or LG, varies based on how far the cable fault extends. The voltage drop across the series resistor is adjusted in accordance with the location of the defect to precisely pinpoint the issue and start the repair process for that particular cable. The fault's precise location is determined using the suggested system. A rectified power supply and an Arduino microcontroller kit are recycled in this structure. Here, the Arduino microcontroller kit and the current detecting circuits, which are manufactured using a variety of resistors, are connected more easily thanks to the integrated ADC device, giving the microcontroller digital data that represents the wire length in kilometers [18].

The group of switches is the source of the malfunction. Distance is shown on a 16x2 LCD display that is connected to the microcontroller. A programmed Arduino microcontroller kit receives the voltage across series resistors in the event of a short circuit, and converts it into precise digital data. This data is then supplied to an ADC to produce a specific fault location in kilometers from the base station [10].

The key objective of the work is to identify and trace the underground cable fault. Instead of running in overhead lines, electrical cables are buried in metropolitan areas. The procedure of fixing a mistake becomes challenging whenever it arises. It is quite problematic to pinpoint the exact position of a flaw in an underground electrical wire line. This research will enable the technical team to address these mistakes more swiftly. Ohm's law serves as the foundation for the system that was constructed here. This along with the short circuit also shows the distance in km where the fault is incurred. Furthermore, it was utilized to show the kind of LCD display failure. When a cable breaks, a buzzer sounds to notify you of the problem and to take quick action.

This method's objectives are to locate the faulty location and determine the distance in kilometers between the base station and the underground cable fault. The basic idea of Current Transformer Theory (CT Theory) is used in this study when a fault like a short circuit happens, the variable current can be determined using a current transformer since the current differs. The length of the cable culpability will determine the voltage drop. This paper proposes an Internet of Things-based

fault location approach for underground power connections. Data will be transmitted via internet access because this model is internet-based. The goals of this method are to identify the faulty spot and using the fundamental ideas of Current Transformer Theory, calculate the distance in kilometers between the base station and the underground cable fault (CT Theory). A current transformer is used to quantify the variable current as it changes during a fault, such as a short circuit. The length of the cable defect will be reflected in the voltage drop across the cable [19].

III. EXISTING SYSTEM

Within the current system, a microcontroller is used to locate the issue in the underground cable. Only the detection of short circuit failures is put forward by this approach. Due to variations in current, the voltage drop resulting from a short circuit or other problem in the cable will vary based on its length.

Thus, the cable is represented by a set of resistors, and one end of the cable is fed a DC voltage. The fault is then observed by measuring the voltage change using an analog to voltage converter, and a microcontroller performs the required calculations to locate the fault distance in kilometers, and display it on an LCD.

A. Drawbacks

The only feature offered by the current technology is the identification of underground cable short circuit faults. The precision of fault location is decreased when 10-bit ADC is used, and accurate data is not obtained. Only the specific distance is used to determine the cable's fault site. Moreover, cables with a voltage range of 1 to 500 kv are supported by the system. It does not use the other cable failure detection method.

The installation of many components may result in an increase in cost of investment.

IV. PROPOSED SYSTEM

The Ohm's law idea is also followed by the proposed system. Here, resistance fluctuation, the identification of short circuit and open circuit faults, and, thanks to technological advancements, the monitoring of the underground cable's temperature indication are all possible. The Arduino Uno, LCD, Wi-Fi modem, and current sensor are the main parts of the system. One of the most crucial tasks is measuring the amount of current. Previously, this was accomplished by adding a resistor to the current routes, but this led to increased heat dissipation. As a result, several technologies have emerged that measure current without causing any heat loss.

Figure 1 shows, When positioned next to a conductor that carries electricity, it responds better and functions at resonant frequency. Here, we have utilized an Arduino, which can be placed over a breadboard due to its small size and read sensed values. It can also perform certain activities based on user-written code. This involves the sensor providing input

to a microcontroller (Arduino), which then uses that input to conduct a number of specified operations, such as displaying data on an LCD display that is interfaced with it or serially transmitting real-time data through the microcontroller.

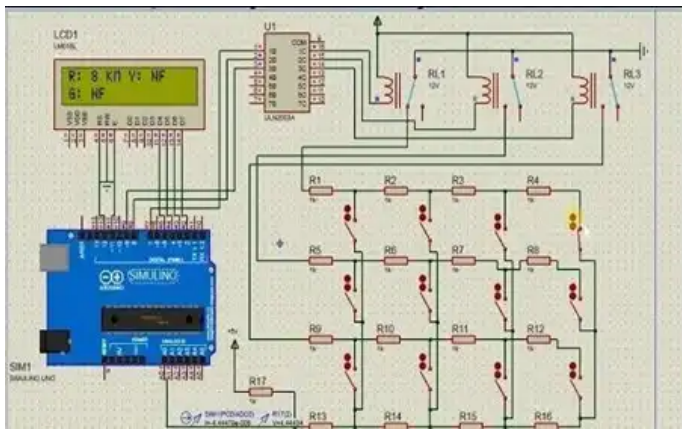


Fig. 1. Circuit Diagram of proposed Model.

V. PROCEDURE

The circuit is made up of an Arduino, a resistance measurement circuit, a power supply, and a 4-line display. Fault switches—roughly twelve switches arranged into three rows of four each—are used to manually induce faults into the kit. The three phases—R, Y, and B—are represented by the rows. There are two positions on each switch: fault (F) and no fault (NF). Low-value resistance measurement is the main part of the underground cable fault detection circuit. It is built using a 100 mAmp continuous current source. In figure 2, The device can monitor very low resistance values because the cables have a resistance of about 0.01 Ohm per meter. A 10-meter cable, for instance, has a resistance of 0.1 Ohm. Resistances up to 50 Ohms can be measured by the circuit. Three series sets of resistances are connected, starting at the reference point. The neutral and the three phases are represented by these sets. Both symmetrical and asymmetrical defects can be found with this technique, including short circuit faults. These three series sets of resistances are used in this study. Over a certain distance, the resistance of the subterranean cable is represented by each series resistor. This configuration uses four resistors in series, which corresponds to a 2–16 km range. Value of each resistance varies.

For each of the phases R, Y, and B, three relays are in use. Each relay's common point is grounded, and the NO points are linked to the three-phase cable. Relay drivers are used to increase the supply and give it to the relays since the supply needed for the relays is greater than what the Arduino can handle.

The system indicates the distance in kilometers where the fault occurred based on the switch operation when a fault is caused by moving any of the 12 switches to the F position. The voltage value changes as a result of the fault and is supplied into the Arduino's ADC by being measured across the

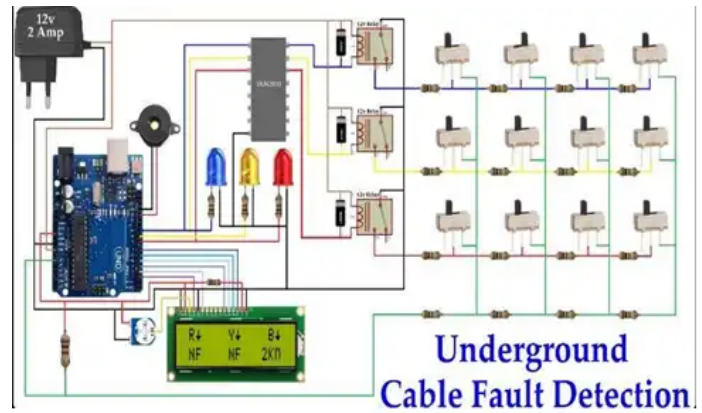


Fig. 2. Block Diagram.

resistance. The Arduino determines the fault's distance using this voltage value.

VI. RESULT ANALYSIS

In this section, the comparative analysis illustrated in Figure 3 highlights the proposed system, which consists of an Arduino kit, switches, a 16x2 LCD, and three LEDs. All these components are powered by a unified power source integrated onto a single board. This work presents the functional model of the proposed system, which will be simulated using the ThinkWorks simulator, with all comparative analyses displayed below. The message "NO FAULT" appears, as seen

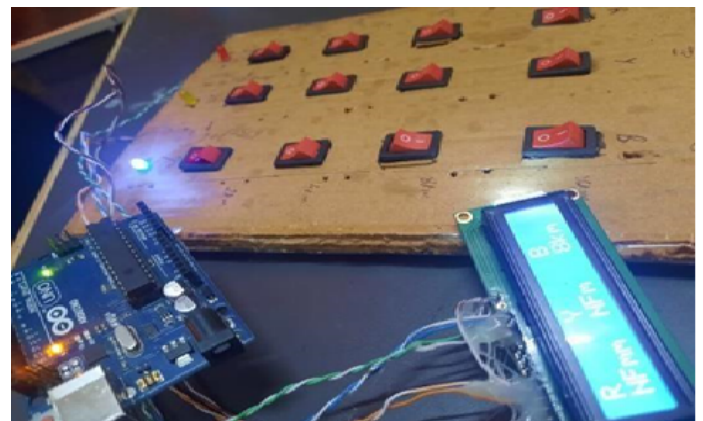


Fig. 3. Circuit Diagram with display.

in the figure 3 whenever the underground cables of the R, Y, and B phases are reported to be functioning properly and no issue persists.

Whenever a fault occurs in the underground cable of the R phase, the message "R phase fault km" for Y phase, the message gets displayed as in the figure 4. An Arduino, a circuit for measuring resistance, a power source, and a 4-line display make up the circuit. Fault switches, which come in three rows of four switches each, are used to manually induce problems into the system. Tabel 1, Table 2 and Table 3 shown The three phases—R, Y, and B—are represented by these rows.

There are two positions on each switch: fault (F) and no fault (NF). Using a 100 mA continuous current source, low-value resistance measurement is the main part of the underground cable failure detection circuit. The device can detect very low resistance values because cables have a resistance of approximately 0.01 Ohm per meter. For example, the resistance of a 10-meter wire is 0.1 Ohm. Because the circuit can measure resistances up to 50 Ohms, it can accommodate different wire lengths. The comparative and data analysis shown in three phase in figure 4, figure 5, figure 6, figure 7 and figure 8.

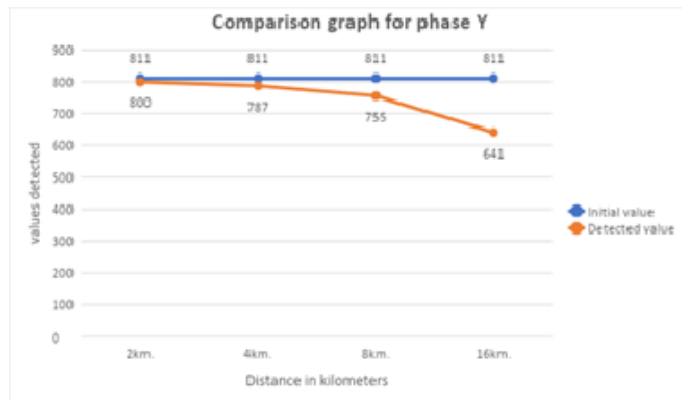


Fig. 4. Comparison for phase Y.

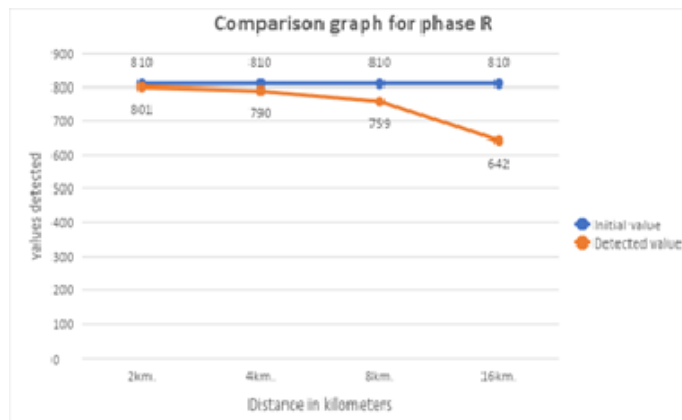


Fig. 5. Comparison for phase R.

CONCLUSION

In conclusion, this work offers a useful and affordable method for utilizing an Arduino-based model to identify defects in subterranean cable systems. The suggested solution uses Ohm's law to precisely determine the distance in kilometers between the base station and the fault, enabling maintenance workers to locate a defect precisely without having to dig up the entire cable. In addition to speeding up repair times, this focused detection technique also drastically lowers labor and repair expenses. This technique presents a timely solution for preserving dependable power distribution and reducing interruptions brought on by cable faults as urban

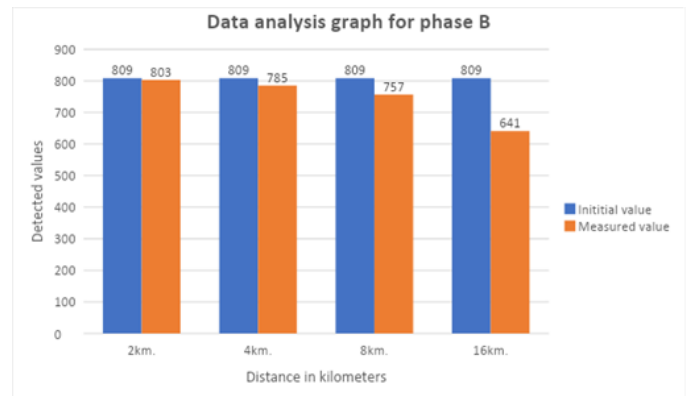


Fig. 6. Data analysis for phase B.

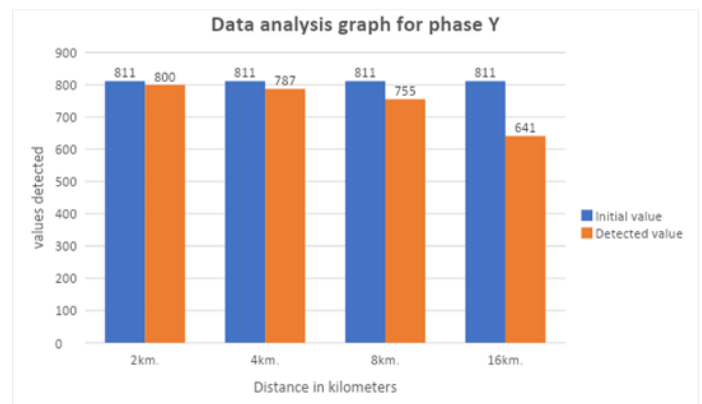


Fig. 7. Data analysis for phase Y

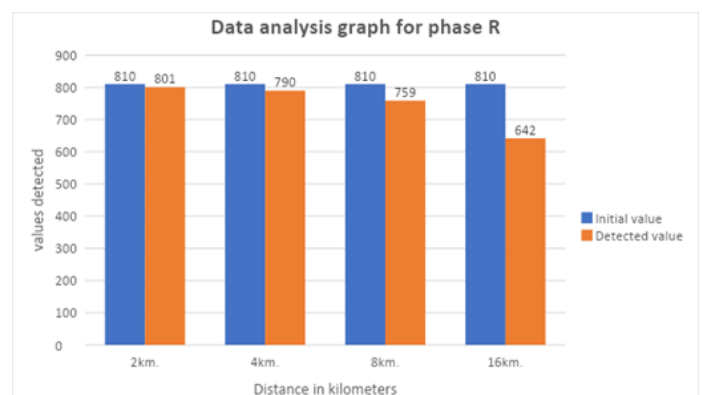


Fig. 8. Data analysis for phase R.

TABLE I
OBSERVATION CHART FOR PHASE VALUES AND PHASE FAULT FOR B PHASE

Initial Value	Value Detected/Measured	Serial Monitor Output/Phase Value	Serial Monitor Output/Phase Fault
809	803	803	2km
809	785	785	4km
809	757	757	8km
809	641	641	16km

TABLE II
OBSERVATION CHART FOR PHASE VALUES AND PHASE FAULT FOR Y PHASE

Initial Value	Value Detected/Measured	Serial Monitor Output/Phase Value	Serial Monitor Output/Phase Fault
811	800	800	2km
811	787	787	4km
811	755	755	8km
811	641	641	16km

TABLE III
OBSERVATION CHART FOR PHASE VALUES AND PHASE FAULT FOR R PHASE

Initial Value	Value Detected/Measured	Serial Monitor Output/Phase Value	Serial Monitor Output/Phase Fault
810	801	801	2km
810	790	790	4km
810	759	759	8km
810	642	642	16km

infrastructure increasingly moves toward underground wiring. Such defect detection devices could eventually be included into contemporary electrical grid management, offering improved subterranean cable resilience and efficiency. This study presents a novel method for identifying faults in subterranean cable systems. This methodology identifies the damaged part, greatly lowering excavation efforts, time, and maintenance costs compared to other methods that require substantial excavation of whole cable lines. The suggested approach improves the effectiveness and dependability of subterranean electrical grid systems by tackling issues like rodent damage and subsurface deterioration, providing an affordable and environmentally friendly substitute for fault detection and repair.

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